Infosys Springboard Virtual Internship 7.0 Milestone 1 Report

Internship Details

Batch	Batch 4
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Internship start date	22 May 2026
Duration	2 Month
Milestone	Data Preparation and Validation

1. Project Title

Cyber Threat Analytics Platform with Incident Visualization System

2. Project Objective

The objective of this project is to prepare enterprise cyber-incident data for reliable analysis and future visualization. The platform is intended to help users explore incident and attack context, detection and response measures, and financial and recovery-related information. Milestone 1 focuses specifically on data preparation: profiling the source CSV, removing exact duplicate records, reviewing data quality, validating fields, assessing unusual numerical values, and adding three derived analytical columns. It does not claim that the interactive dashboard or predictive analytics system is already complete.

3. Tools and Technologies

- **Dataset:** Enterprise Cyber Kill Chain dataset, supplied as a CSV file.
- **Source file:** `Enterprise_Cyber_Kill_Chain_Dataset(2).csv`
- **Microsoft Power BI Desktop:** Used to import the CSV dataset, prepare and validate data, create derived columns, and support analysis and visualization. Dashboard development is part of the subsequent project work.

4. Project Execution

The project workflow below summarizes the planned end-to-end process for the Cyber Threat Analytics Platform with Incident Visualization System, from defining the problem and preparing the dataset to report deployment. For Milestone 1, the primary focus was on dataset collection, understanding, preprocessing, and validation; later analytical and dashboard stages represent the broader workflow.

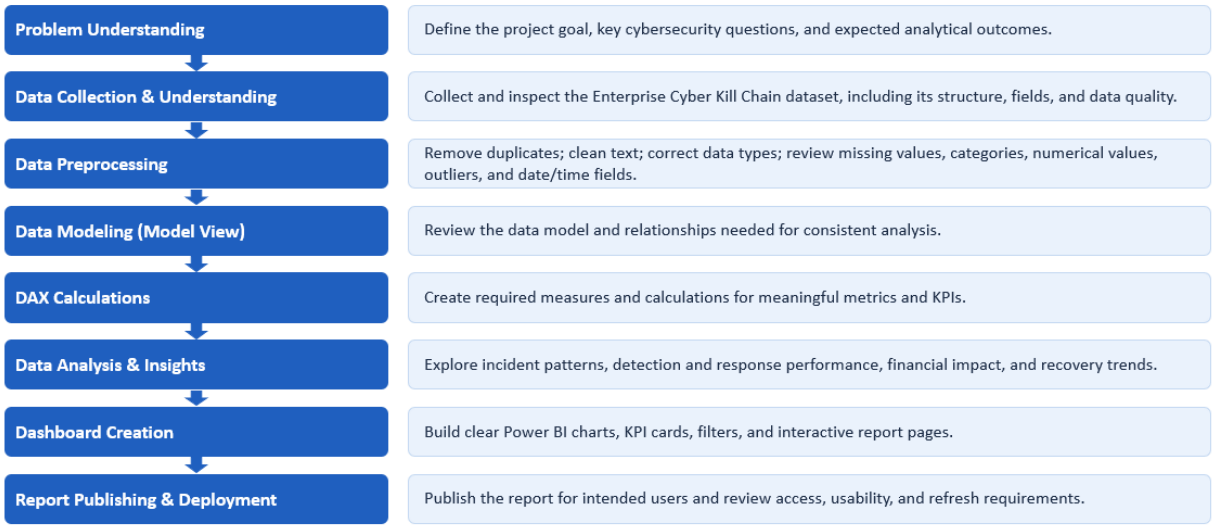


Figure 1: Power BI workflow for the Cyber Threat Analytics Platform with Incident Visualization System.

5. Timeline Overview

The timeline below separates the data-preparation work described for Milestone 1 from planned future work. The table records the two activities documented for this report; additional activities can be added when confirmed.

Week	Planned activity	Status / outcome
Week 1	Dataset exploration and profiling	Source profiled: 100,500 rows and 50 columns; 500 exact duplicate rows removed.
Week 2	Validation, missing-value review, outlier assessment and feature creation	Milestone 1 preparation activities described in this report; three derived columns created.

6. Key Milestones

Milestone	Description	Date
Project kickoff	Dataset selected and initial scope established.	22 May 2026
Milestone 1 / first draft	Dataset cleaning and validation activities completed; final prepared structure is 100,000 rows by 53 columns, including three derived fields.	To be confirmed

Data Preprocessing Contents

The following contents page provides a quick guide to the data-preprocessing section of Milestone 1.

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7. Detailed Data Preprocessing Steps

The preparation process focused on improving the consistency and analytical readiness of the dataset. The steps below describe the checks and transformations performed, without assuming unprovided formulas, thresholds, counts of missing values, or outlier statistics.

6.1 Dataset inspection and profiling

- Loaded the source file `Enterprise_Cyber_Kill_Chain_Dataset(2).csv` into Microsoft Power BI Desktop for data preparation and review.
- Reviewed the overall dataset structure, including its row count, column count, column names and sample records.
- Grouped the columns conceptually into six areas: incident and attack details; detection metrics; response metrics; financial impact; recovery and remediation; and time and incident context.
- Used this initial review to identify fields requiring type checks, categorical consistency checks, missing-value review and numerical validation.

Incident_ID	Timestamp	Industry	Country	Company_Size	Employee_Count	Attack_Vector	Threat_Actor	Firewall	MFA	EDR	IDS	Security_Training	Password_
8115	26-03-2023 12:30:00	Retail	India	Small	158	Email	Hacktivist	1	1	1	1	1	Moderate
86697	22-06-2025 02:00:00	Banking	Australia	Small	65	VPN	Cyber Criminal	1	1	1	1	1	Moderate
88700	12-07-2025 22:45:00	Manufacturing	Singapore	Small	67	Cloud	Cyber Criminal	1	1	1	1	1	Moderate
72939	29-01-2025 18:30:00	Technology	Canada	Small	136	Web	Cyber Criminal	1	1	1	1	1	Strong
21711	15-08-2023 03:30:00	Manufacturing	Canada	Small	247	USB	Cyber Criminal	1	1	1	1	1	Strong
65591	14-11-2024 05:30:00	Banking	USA	Small	161	Cloud	Cyber Criminal	1	1	1	1	1	Strong
94677	13-09-2025 05:00:00	Retail	Japan	Small	66	API	Cyber Criminal	1	1	1	1	1	Moderate
71555	15-01-2025 08:30:00	Telecom	UK	Small	209	Web	Cyber Criminal	1	1	1	1	1	Moderate
75754	28-02-2025 02:15:00	Technology	India	Small	20	Email	Hacktivist	1	1	1	1	1	Strong
35010	31-12-2023 16:15:00	Retail	India	Small	31	Email	Cyber Criminal	1	1	1	1	1	Moderate
41441	07-03-2024 16:00:00	Manufacturing	Singapore	Small	129	Email	Cyber Criminal	1	1	1	1	1	Moderate
95333	20-09-2025 01:00:00	Healthcare	Canada	Small	183	Web	Cyber Criminal	1	1	1	1	1	Moderate
60486	22-09-2024 01:15:00	Healthcare	UK	Small	80	Email	Cyber Criminal	1	1	1	1	1	Moderate
64001	28-10-2024 16:00:00	Banking	Singapore	Small	71	Cloud	Cyber Criminal	1	1	1	1	1	Strong
97438	11-10-2025 23:15:00	Healthcare	Australia	Small	55	Email	Cyber Criminal	1	1	1	1	1	Strong
57381	20-08-2024 17:00:00	Banking	Japan	Small	184	Web	Cyber Criminal	1	1	1	1	1	Moderate
86793	23-06-2025 02:00:00	Manufacturing	USA	Small	58	Email	Cyber Criminal	1	1	1	1	1	Weak
46520	29-04-2024 13:45:00	Healthcare	Germany	Small	182	USB	Cyber Criminal	1	1	1	1	1	Moderate
13958	26-05-2023 09:15:00	Retail	USA	Small	77	Web	Cyber Criminal	1	1	1	1	1	Moderate
82078	04-05-2025 23:15:00	Technology	Canada	Small	29	VPN	Cyber Criminal	1	1	1	1	1	Strong
48345	18-05-2024 14:00:00	Manufacturing	Singapore	Small	126	Cloud	Cyber Criminal	1	1	1	1	1	Strong
24037	08-09-2023 09:00:00	Banking	Germany	Small	69	API	Hacktivist	1	1	1	1	1	Moderate
56720	13-08-2024 19:45:00	Retail	USA	Small	226	Email	Cyber Criminal	1	1	1	1	1	Moderate
99386	01-11-2025 06:15:00	Retail	Germany	Small	227	Web	Cyber Criminal	1	1	1	1	1	Strong
54664	23-07-2024 09:45:00	Retail	Germany	Small	44	VPN	Hacktivist	1	1	1	1	1	Strong
3077	02-02-2023 01:00:00	Telecom	Japan	Small	181	API	Cyber Criminal	1	1	1	1	1	Strong
38687	07-02-2024 23:30:00	Banking	UK	Small	237	USB	Cyber Criminal	1	1	1	1	1	Moderate

Figure 2: Dataset inspection and profiling of the source CSV.

6.2 Exact duplicate removal

- Checked for exact duplicate records across all columns, rather than removing records based on a single field.
- Removed 500 exact duplicate rows while retaining one copy of each repeated record.
- The row count therefore changed from 100,500 to 100,000.
- The original 50 columns were retained at this stage; duplicate removal changed the number of rows, not the column schema.

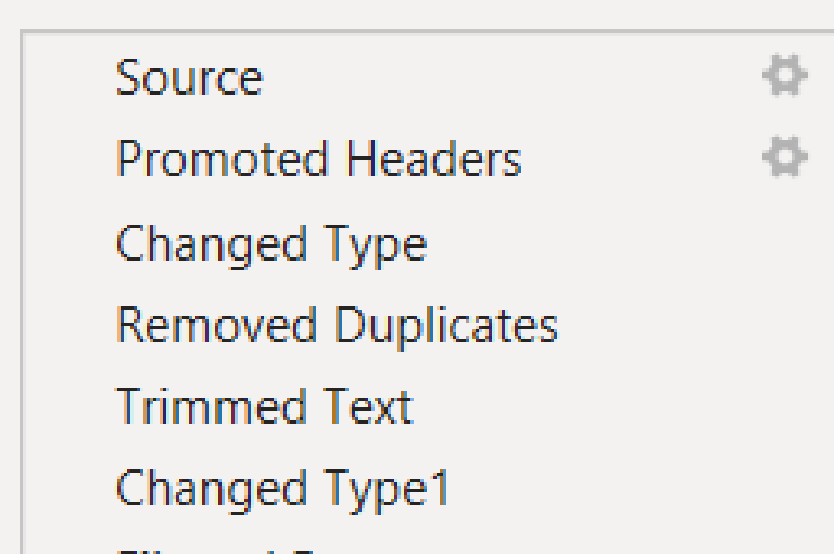


Figure 3: Removal of 500 exact duplicate rows across all columns.

6.3 Numerical, categorical and consistency validation

- **Numerical validation:** Reviewed relevant numerical fields for invalid or unsuitable values and checked whether values were appropriate for their intended meaning.
- **Categorical validation:** Reviewed category values for consistency, including text values that should use consistent labels or formatting.
- **Data consistency:** Checked relevant fields for contradictory or inconsistent entries and reviewed date/time consistency where applicable.
- These checks were intended to improve reliability before analysis; no unsupported validation totals or pass rates are reported.

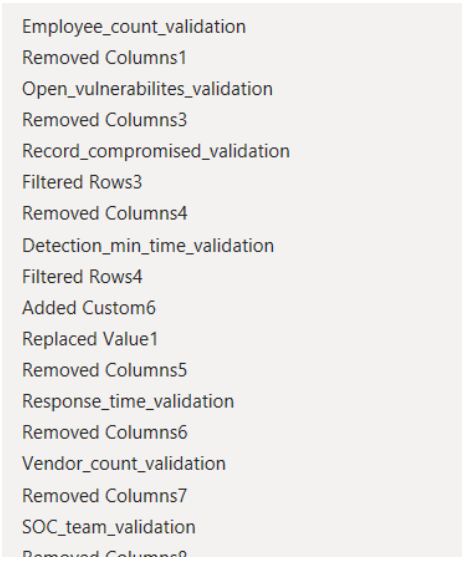


Figure 4: Numerical, categorical and data-consistency validation.

6.4 Data type checking

- Reviewed data types for relevant columns so that numeric measures, categorical labels and time-related fields could be interpreted appropriately.
- Considered whether the field's content matched its assigned type and whether the type was suitable for subsequent analysis.
- Type checking was treated as a validation step; this report does not claim that every column was converted or that a particular conversion was applied where no evidence was supplied.

6.5 Missing-value review

- Reviewed missing or unknown values in the dataset.
- Did **not** automatically replace unknown values with zero. A missing value means that information is unavailable, whereas zero can be a valid measured value.
- This approach avoids introducing false information into later summaries and comparisons.
- No missing-value counts or imputation statistics are stated because those values were not provided.

6.6 Outlier assessment

- Assessed unusual numerical values in context as part of the validation workflow.
- Plausible numerical outliers were retained rather than automatically deleted, since an unusually high or low value can represent a genuine incident.
- The report does not assume a specific detection algorithm, statistical threshold, lower or upper bound, or number of outliers.

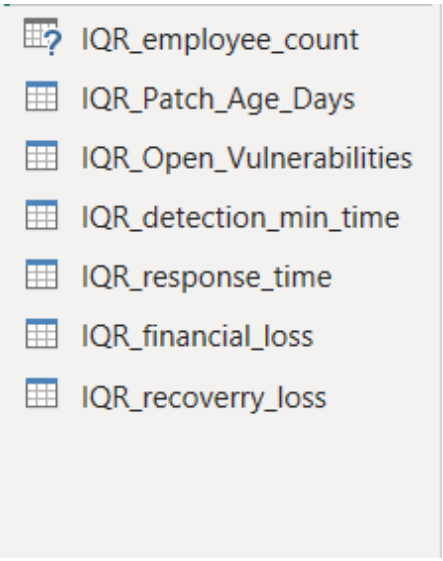


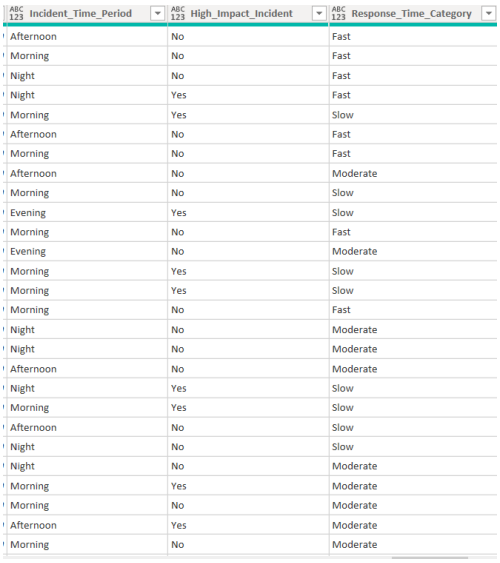
Figure 5: Assessment of unusual numerical values; plausible outliers were retained.

6.7 Feature creation Three derived columns were added to make selected incident attributes easier to interpret:

- `Incident_Time_Period`: Represents the time-of-day period associated with an incident. Values visible in the supplied screenshot include Morning, Afternoon, Evening and Night.

- `High_Impact_Incident`: A Yes/No indicator for whether an incident is classified as high impact.
- `Response_Time_Category`: Groups response time into readable categories. Values visible in the supplied screenshot include Fast, Moderate and Slow.

These descriptions reflect the visible field names and sample category labels. Specific formulas, thresholds and classification rules are not documented here because they were not supplied.



Incident_Time_Period	High_Impact_Incident	Response_Time_Category
Afternoon	No	Fast
Morning	No	Fast
Night	No	Fast
Night	Yes	Fast
Morning	Yes	Slow
Afternoon	No	Fast
Morning	No	Fast
Afternoon	No	Moderate
Morning	No	Slow
Evening	Yes	Slow
Morning	No	Fast
Evening	No	Moderate
Morning	Yes	Slow
Morning	Yes	Slow
Morning	No	Fast
Night	No	Moderate
Night	No	Moderate
Afternoon	No	Moderate
Night	Yes	Slow
Morning	Yes	Slow
Afternoon	No	Slow
Night	No	Slow
Night	No	Moderate
Morning	Yes	Moderate
Morning	No	Moderate
Afternoon	Yes	Moderate
Morning	No	Moderate

Figure 6: The three derived analytical fields added during feature creation.

6.8 Final structure review

- Rows after duplicate removal: 100,000.
- Original columns retained: 50.
- Derived columns added: 3.
- Final prepared structure: **100,000 rows** $\times$ **53 columns**.

8. Dataset Overview and Column Groups

The dataset is named *Enterprise Cyber Kill Chain*. Its source file is `Enterprise_Cyber_Kill_Chain_Dataset(2).csv`.

Dataset measure	Value
Original rows	100,500
Original columns	50
Exact duplicate rows removed	500
Rows after duplicate removal	100,000
Derived analytical columns added	3
Final prepared structure	100,000 rows × 53 columns

Six high-level column groups

1. **Incident and attack details:** Incident-related information and attack attributes.
2. **Detection metrics:** Detection time and related detection measures, including `Detection_Time_Min` where present.
3. **Response metrics:** Response time and response-related fields, including `Response_Time_Min` and `Response_Time_Category`.
4. **Financial impact:** Financial-loss information and impact indicators, including `Financial_Loss_USD` where present.
5. **Recovery and remediation:** Recovery-cost and recovery-related fields, including `Recovery_Cost_USD` where present.
6. **Time and incident context:** Timestamps, incident time-of-day groupings and related context, including `Timestamp` and `Incident_Time_Period` where present.

9. Challenges and Resolutions

Challenge	Resolution
Exact duplicate records	Checked complete rows across all columns and removed 500 exact duplicates.
Missing or unknown values	Reviewed missing values without automatically treating them as zero.
Inconsistent category text	Reviewed categorical consistency and cleaned selected text fields where required.
Different data types	Reviewed suitable types for relevant fields and checked values against their intended meaning.
Unusual numerical values	Assessed outliers in context and retained plausible values rather than deleting them automatically.
Interpreting raw incident attributes	Added `Incident\_Time\_Period`, `High\_Impact\_Incident` and `Response\_Time\_Category`.

10. Outcome of Milestone 1

- Removed 500 exact duplicate rows, reducing the dataset from 100,500 to 100,000 rows.
- Reviewed numerical values, categorical consistency, data types, missing values and date/time consistency.
- Assessed unusual numerical values and retained plausible outliers rather than removing them automatically.
- Added three derived columns to improve the readability of incident time, impact and response speed.
- Prepared a structured dataset of 100,000 rows and 53 columns for future analysis.

Milestone 1 established a foundation for future cybersecurity incident analytics through data preparation, validation and feature creation. The interactive Power BI dashboard and any predictive analytics remain future work.

11. Future Work

- Develop interactive Power BI dashboards using the prepared dataset.
- Visualize cyber incident patterns and explore incident context.
- Analyse detection and response measures.
- Explore financial impact and recovery-cost information.
- Consider advanced or predictive analytics if the dataset and project scope support it.
- Evaluate potential real-time integration or cloud deployment as a future option.

Conclusion

The *Cyber Threat Analytics Platform with Incident Visualization System* aims to transform raw cybersecurity incident data into meaningful insights for identifying attack patterns, evaluating detection and response performance, and understanding the financial impact of cyber incidents.

In Milestone 1, the Enterprise Cyber Kill Chain dataset was prepared for further analysis through duplicate removal, data validation, data type review, missing-value assessment, outlier evaluation, and feature creation. A total of 500 exact duplicate rows were removed, reducing the dataset from 100,500 to 100,000 rows. Three derived columns were added: `Incident\_Time\_Period`, `High\_Impact\_Incident`, and `Response\_Time\_Category`, resulting in a prepared dataset containing 100,000 rows and 53 columns.

This milestone establishes a foundation for the next stages of the project. Future work will focus on developing interactive Power BI dashboards to visualize incident trends, detection and response metrics, financial impact, and recovery-related information. These visual insights are intended to support better understanding of cybersecurity incidents and more informed decision-making.

Acknowledgements

We would like to express our heartfelt gratitude to the **Infosys Springboard Team** for organizing this valuable internship opportunity and providing a structured platform for learning and professional development.

We are sincerely thankful to our **mentor** for her continuous guidance, constructive feedback, and encouragement throughout the project. Her valuable suggestions helped us improve our analytical approach and execute the project effectively.

We extend our appreciation to **our teammates** for their collaboration, mutual support, idea sharing, and dedicated teamwork during every phase of the project. We are also grateful for the opportunity to work with a real-world dataset, which enabled us to apply theoretical analytical concepts in a practical environment.

Working with **Power BI, data preprocessing, data modeling, and interactive dashboard development** significantly enhanced our technical knowledge and problem-solving abilities. Furthermore, the collaborative environment, regular reviews, and milestone-based execution helped us develop essential professional skills, including communication, accountability, time management, and coordination.

This internship has been a valuable learning experience and an important milestone in our professional growth, strengthening our technical expertise and preparing us for future challenges in data analytics.

12. References

- **Dataset:** Enterprise Cyber Kill Chain dataset. Source file: `Enterprise_Cyber_Kill_Chain_Dataset(2).csv`.
- **Microsoft Power BI Desktop documentation:** https://learn.microsoft.com/power-bi/fundamentals/desktop-what-is-desktop